Md Tawshif Islam Nahian

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Summary — Computer Science and Engineering graduate combining production software engineering with applied AI development, backed by a research record in causal inference, graph neural networks, and Bayesian optimization, including a 1st Best Paper Award at SPECTRA 2026. Worked on and developed core banking modules in Java at a fintech firm, then moved into building production AI agents for an EdTech platform: automated grading, adaptive tutoring, exam evaluation. Pairing that enterprise background with the numerical and algorithmic problem-solving developed through applied research in causal inference and optimization ties the two sides together.

EDUCATION

Rajshahi University of Engineering and Technology (RUET), Bangladesh Feb 2019 – Aug 2025
B.Sc. in Computer Science and Engineering (CGPA: 2.81/4.00)

Kushtia Government College, Bangladesh May 2016 – Apr 2018
Higher Secondary Certificate, Science (GPA: 5.00/5.00)

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C/C++
Backend & Enterprise Systems: Spring Boot, RESTful APIs, TAFJ, Temenos T24 Core Banking
Databases & Querying: SQL, MySQL, MongoDB, SQLite, H2, Temenos T24 NOFILE Enquiries
Machine Learning & Applied AI: PyTorch, Scikit-learn, XGBoost, Graph Neural Networks (GNNs), CNNs, SHAP, imbalanced-learn, NumPy, Pandas, statsmodels
Computational & Numerical Methods: Causal Discovery (PC Algorithm, NOTEARS), Bayesian Optimization (Gaussian Processes), Mediation Analysis, Class-Imbalance Correction (SMOTE variants)
Computer Vision & NLP: OpenCV, Hugging Face, Large Language Models (LLMs)
AI Agent Frameworks: LangChain
Algorithmic & Cognitive Modeling: Item Response Theory (IRT), Semantic Similarity Scoring
Research & Development Tools: Git, Streamlit, LaTeX, Postman, Linux/Unix

WORK EXPERIENCE

BongoBrain IT, Remote Jun 2026 – Present
AI Engineer [Certificate]

- Working on an OMR sheet-scanning agent that grades multiple-choice answers and writes out the correct solution wherever a student gets one wrong.
- Building part of a student-progress tracker that figures out what level a student is actually at — say, beginner-level vectors versus mid-level Newtonian mechanics — and adjusts the next question to match.
- Co-building an exam-evaluation agent that scores open-ended answers with NLP instead of simple keyword matching.
- Coded a module for an internal accounts agent that checks submitted receipts against a set of rules and flags them accept or reject.
- Currently gathering insulin-intake and glucose-management data as the first step toward a future diabetic-care prediction project, with model development still ahead.

Nazihar IT Solutions Limited, Dhaka, Bangladesh Dec 2025 – May 2026
Software Engineer Intern [Certificate]

- Contributed to the Temenos T24 R22 core banking reimplementation at JBPLC, the country's third-largest bank, on the CBS development track.
- Wrote T24 FILE and NOFILE Enquiry routines with Java for high-throughput ledger data retrieval, spanning the Accounts, Customer, and Trade Finance modules.

- Built T24 Version routines and transactional API routines for Trade Finance (TF) module development.
- Explored Jython to cut down routine length and translated existing InfoBasic routines into Java, picking up the underlying banking logic along the way.
- Studied the banking logic embedded in the Temenos environment to build a clearer picture of the underlying business processes.

Outlier.ai, Remote Mar 2023 – Nov 2024

Quality Assurance Tasker — AI Quality & Validation

- Investigated system outputs and conducted root-cause analysis to identify AI model errors and improve operational quality.
- Applied structured QA principles to verify logical consistency and produced structured analytical reports.

MACHINE LEARNING PROJECTS

- **PGCB-Forecasting:** Built an end-to-end load-forecasting pipeline using XGBoost and LSTM with 72-hour recursive predictions and 95% confidence intervals. [GitHub]

RESEARCH EXPERIENCE

Independent Research 2024 – Present

Focus: Causal Inference & Adaptive Optimization

- **Feature-Wise Adaptive Imputation(FAI)**: Built a per-feature imputation selector — six statistical descriptors (skewness, variance, entropy, correlation) feeding a Random Forest — that stayed within 1% of the best baseline across 9 missingness conditions.
- **Causal Bias-Decomposition(A-ICF):** Coded a causal-discovery and mediation-decomposition pipeline (TV = SE + IE − DE) with constrained optimization and bootstrapped confidence intervals; an ablation traced the real accuracy bottleneck to the encoding stage, not the fairness penalty.
- **Regret-Minimizing Adaptive Optimization(RMA-BO):** Derived a rule for a Gaussian Process optimizer that adapts its exploration parameter by scoring acquisition regret over a sliding window, validated across synthetic and real-world benchmarks.

Collaboration 2024 – present

Focus: Fairness-Aware & Trustworthy Machine Learning

- **Equity Graph Attention Network (CF-EGAT):** Co-built a graph attention network folding causal-influence and fairness terms into the attention weights, with a Dirichlet uncertainty head — best calibration of any model tested. *(1st Best Paper, SPECTRA 2026)*
- **Crime Classification & DemocracyGuard**: Ran nested cross-validation pipelines (XGBoost + SMOTE; ensemble forecasting) on crime-classification and democratic-backsliding data, reporting SHAP attributions and bootstrapped calibration metrics.
- **Multi-Objective Trust Auditing(ANCI):** Scored neurons by gradient overlap between task and fairness loss, then restricted multi-objective regularization to that subset, validated via 5-seed ablations across fairness, robustness, and privacy.

Final Year Thesis, RUET Apr 2025 – Aug 2025

Focus: Deep Learning for Medical Image Analysis

- **Deep CNN For Cancer Detection:** Trained CNN classifiers on chest histopathological images to detect malignant tissue, using augmentation and transfer learning for a small labeled dataset.

PUBLICATIONS & SCHOLARLY MANUSCRIPTS

- Co-authored, CF-EGAT: A Causal Fairness-Aware Equity Graph Attention Network for Country-Level Environmental Livability Classification - *SPECTRA 2026* (Oral Presentation, 1st Best Paper Award). DOI: 10.5281/zenodo.21195761.
- Co-authored, RMA-BO: Regret-Minimizing Adaptive Bayesian Optimization - *SPECTRA 2026* (Oral Presentation). DOI: 10.5281/zenodo.21194394.
- Co-authored, Machine Learning for Crime Classification: A Fairness-Aware Approach to Class Imbalance - *Journal of Machine Learning Applications (JMLA)*, 2(1), 9–17. DOI: 10.61577/jmla.2026.100002.

- Co-authored, DemocracyGuard: Testing a Divergence-Index Reconciliation of Subjective and Objective Democracy Indicators for Forecasting Adverse Regime Transitions. *Under Review at Transactions on Machine Learning Research (TMLR)*.
- Co-authored, Auditing, Not Predicting: A Causal Bias-Decomposition Framework for Clinical Fairness, and a Cautionary Tale of Representation Collapse in A-ICF. *Under Review at IEEE OMLET 2026*.
- Co-authored, Active Neuro-Causal Immunization (ANCI): A Conflict-Neuron Framework for Jointly Auditing Fairness, Robustness, and Privacy in Neural Classifiers. *Under Review at ICRPSET 2026*
- Co-authored, FAI: Feature-Wise Adaptive Imputation via Downstream-Aware Method Selection. *Under Review at ICISET 2026*.

AWARDS & ACCOLADES

- 1st Best Paper Award, SPECTRA 2026 for "CF-EGAT" (see Publications). [Certificate]
- Government of Bangladesh Merit Scholarship for top academic distinction, SSC 2016 & HSC 2018.
- 1st Place, Creative Writing Competition, Bangladesh Agricultural University Cultural Club (BAUCC).

EXTRACURRICULAR ACTIVITIES & HOBBIES

- **Short Film Projects :** Wrote the screenplay for and co-directed two short films titled "Opor Prishtha"(Next Page) and "Odoito"(Non-Duality). Links : Next Page . Non-Duality
- **Creative Writing:** Published hundreds of articles and poems on social media; placed 1st in a creative writing competition.
- **Reading & Film:** Avid reader of novels and fiction with a strong interest in cinema.
- **Trekking:** Regular mountain and trail hiking.

LEADERSHIP & VOLUNTEERING

- **Student Mentorship(2019-present) :** Mentored junior engineering students in programming and core STEM subjects.
- **Community & Youth Outreach(2010-present):** Organized and delivered community outreach and youth engagement initiatives through local cultural programs.